



CAMPUS LINK CRM

PROJECT DOCUMENTATION

Problem Statement

Universities receive hundreds of academic and administrative queries from students each semester. Currently, these queries are managed manually via emails or paper forms, which results in:

- Delays in response and resolution
- Missed follow-ups
- Lack of visibility into resolution timelines

Additionally, student academic records are stored in scattered systems, making it difficult for faculty and administrators to quickly generate actionable insights.

Phase 1: Problem Understanding & Industry Analysis

Objective:

To understand the problems faced by universities in managing student queries and academic records, analyze industry practices, and define the initial direction for the Campus Connect Salesforce implementation.

Tasks Performed:

1. Requirement Gathering

- Collected pain points from students (delayed responses, lack of transparency).
- Gathered feedback from faculty and administration on challenges in query resolution and academic tracking.
- Defined high-level needs: query management, automation, dashboards, and centralized student profiles.

2. Stakeholder Analysis

- Students: Need a simple way to log queries and track progress.
- Faculty: Require visibility into student issues and course-related queries.

- Administration: Need control over non-academic queries (finance, facilities) and overall monitoring.
- IT Team: Responsible for CRM setup, automation, and maintenance.

3. Business Process Mapping

- Current flow: Student → Email/Form → Manual forwarding → Faculty/Admin → Delayed resolution.
- Desired flow: Student → Salesforce Query → Auto Assignment → Faculty/Admin → Real-time Tracking via Dashboards.

4. Industry-Specific Use Case Analysis

- Benchmarked existing university CRM practices.
- Identified common patterns: centralized query desks, case tracking, and mobile-first approaches.
- Highlighted Salesforce advantages: automation, scalability, and analytics.

5. AppExchange Exploration

- Explored education-related Salesforce solutions on AppExchange.
- Examples: Salesforce Education Cloud, Student Success Hub.
- Found reusable patterns like case management templates and student lifecycle apps.

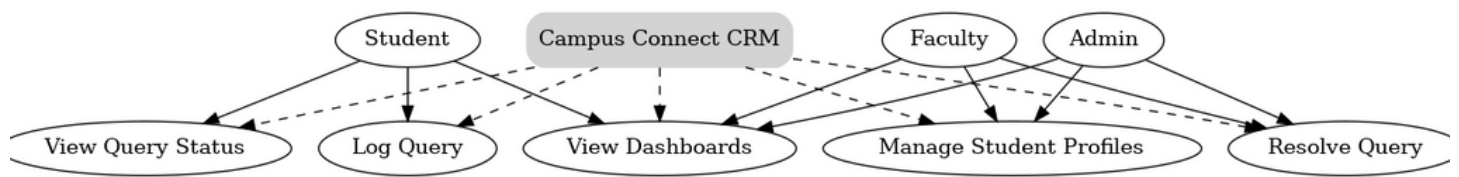
✅ Functional Requirements:

1. Centralized query logging system.
2. Automated assignment of queries to departments (Academic, Finance, Administration).
3. Student profile with basic details (Name, Email, Enrollment, Course mapping).
4. Dashboards for:
 - Students (query status)
 - Faculty (open/resolved queries)
 - Admin (overall query analytics).

🔒 Non-Functional Requirements:

1. **Performance:** Query creation should take less than 3 seconds.
2. **Security:** Student data must be restricted to authorized users only.
3. **Scalability:** Should handle queries from 10,000+ students.
4. **Usability:** Mobile-friendly UI (for students on smartphones).

Use Case Diagram:



Phase 2 – Org Setup & Configuration: CAMPUS LINK Project

Phase Goal:

Configure Salesforce Org for the CAMPUS LINK Portal to support workflow for Students, Faculty, and Department Heads. Establish all users, roles, profiles, permission sets, and security settings tailored to the campus query management scenario.

Salesforce Edition Used:

- Salesforce Developer Edition
- Supports custom objects, Apex, LWC, and Flows for advanced workflow.

Company Information Setup

- Institute/Organization Name: Set to official campus/institution branding (ie. Gyan Ganga Institute of Technology and Sciences).
- Business Hours: Defined as 9 AM to 6 PM, Monday to Friday, reflecting standard academic schedules.
- Official Holidays: Entered to block non-operational dates and avoid query escalations on off-days.
- Fiscal Year: Retained standard fiscal year unless campus has unique requirements.

User Setup & Licensing

- Student Users:
 - Created as Salesforce Users for demo purpose (in real world will be created as Community Users or External users).

- Assigned licenses: Salesforce.
- Each mapped to a Student__c record for full linking with query workflow.

User Detail [Edit](#) [Sharing](#) [Reset Password](#) [Login](#) [Freeze](#) [View Summary](#)

Name	Test student3	Role	Student - P
Alias	depa	User License	Salesforce
Email	sahib04karan03@gmail.com [Verified]	Profile	Student Profile
Username	jassimanager1@gmail.com	Active	☒
Nickname	User17543996510185163059 i	Marketing User	☐
Title		Offline User	☐
Company		Knowledge User	☐
Department		Flow User	☐
Division		Service Cloud User	☐
Address	India	Site.com Contributor User	☐
Time Zone	(GMT-07:00) Pacific Daylight Time (America/Los_Angeles)	Site.com Publisher User	☐
Locale	English (United States)	WDC User	☐
Language	English	Mobile Push Registrations	View
Delegated Approver		Data.com User Type	i
Manager		Accessibility Mode (Classic Only)	☐ i
Receive Approval Request Emails	Only if I am an approver	Debug Mode	☐ i
Federation ID		High-Contrast Palette on Charts	☐ i
App Registration: One-Time Password Authenticator	i	Load Lightning Pages While Scrolling	☒ i
App Registration: Salesforce Authenticator	i	Salesforce CRM Content User	☒
Security Key (U2F or WebAuthn)	i	Receive Salesforce CRM Content Email Alerts	☒
Lightning Login	i	Receive Salesforce CRM Content Alerts as Daily Digest	☒
Temporary Verification Code (Expires in 1 to 24 Hours)	[Generate] i	Make Setup My Default Landing Page	☐
		Allow Forecasting	☐
		No MRU Updates	☐ i
		Call Center	

- Faculty Users:
 - Created as internal Salesforce users.
 - Assigned Salesforce Platform licenses as required.
 - Each mapped to a Faculty__c record.

SETUP
Users

User
Faculty

[Permission Set Assignments \(1\)](#) | [Permission Set Assignments: Activation Required \(0\)](#) | [Permission Set Group Assignments \(0\)](#) | [Permission Set License Assignments \(0\)](#) | [Personal Groups \(0\)](#) | [Public Group Memberships \(0\)](#) | [Queue Memberships \(0\)](#) | [Third-Party Account Links \(0\)](#) | [Built-in Authenticators \(0\)](#) | [Installed Mobile Apps \(0\)](#) | [Authentication Settings for External Systems \(0\)](#) | [Login History \(0+\)](#) | [User Provisioning Accounts](#)

User Detail [Edit](#) [Sharing](#) [Reset Password](#) [Login](#) [Freeze](#) [View Summary](#)

Name	Faculty	Role	Faculty - CSE
Alias	tsr1	User License	Salesforce Platform
Email	jassirep1@gmail.com [Verify] i	Profile	Standard Platform User
Username	rep1@ggits.com	Active	☒
Nickname	User1757532030073410707 i	Marketing User	☐
Title		Offline User	☐
Company		Knowledge User	☐
Department		Flow User	☐
Division		Service Cloud User	☐
Address		Site.com Contributor User	☐
Time Zone	(GMT+05:30) India Standard Time (Asia/Kolkata)	Site.com Publisher User	☐
Locale	English (United States)	WDC User	☐
Language	English	Mobile Push Registrations	View
Delegated Approver		Data.com User Type	i
Manager		Accessibility Mode (Classic Only)	☐ i
Receive Approval Request Emails	Only if I am an approver	Debug Mode	☐ i
Federation ID		High-Contrast Palette on Charts	☐ i
App Registration: One-Time Password Authenticator	i	Load Lightning Pages While Scrolling	☒ i
App Registration: Salesforce Authenticator	i	Salesforce CRM Content User	☐
Security Key (U2F or WebAuthn)	i	Receive Salesforce CRM Content Email Alerts	☐
Lightning Login	i	Receive Salesforce CRM Content Alerts as Daily Digest	☐
Temporary Verification Code (Expires in 1 to 24 Hours)	[Generate] i	Make Setup My Default Landing Page	☐

- Department Head Users:
 - Created to manage and assign queries across departments.
 - Assigned Salesforce license for broader access.

SETUP

Users

User

Department_Head

[Permission Set Assignments \(1\)](#) | [Permission Set Assignments: Activation Required \(0\)](#) | [Permission Set Group Assignments \(0\)](#) | [Permission Set License Assignments \(0\)](#) | [Personal Groups \(0\)](#) | [Public Group Membership \(0\)](#) | [Queue Membership \(0\)](#) | [Third-Party Account Links \(0\)](#) | [Built-in Authenticators \(0\)](#) | [Installed Mobile Apps \(0\)](#) | [Authentication Settings for External Systems \(0\)](#) | [Login History \(14\)](#) | [User Provisioning Accounts \(1\)](#)

User Detail

Edit

Sharing

Reset Password

Login

Freeze

View Summary

Name	Department_Head	Role	Department Head - IOT
Alias	tsr2	User License	Salesforce
Email	jassimanager1@gmail.com (Verify) i	Profile	Department Head Profile
Username	manager1@ggits.com	Active	☒
Nickname	User17575334245943688255 i	Marketing User	☐
Title		Offline User	☐
Company		Knowledge User	☐
Department		Flow User	☐
Division		Service Cloud User	☐
Address		Site.com Contributor User	☐
Time Zone	(GMT+05:30) India Standard Time (Asia/Kolkata)	Site.com Publisher User	☐
Locale	English (United States)	WDC User	☐
Language	English	Mobile Push Registrations	View
Delegated Approver		Data.com User Type	i
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Lightning Login	i	Receive Salesforce CRM Content Alerts as Daily Digest	☒
Temporary Verification Code (Expires in 1 to 24 Hours)	(Generate) i	Make Setup My Default Landing Page	☐

Profiles and Permissions

- Custom Profiles Cloned from Standard User:
 - Student Profile: Read access to Student, Create/Read own Student Query. No access to assign/resolution fields.

SETUP

Profiles

Profile

Student Profile

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

[Login IP Ranges \(0\)](#) | [Enabled Apex Class Access \(1\)](#) | [Enabled Visualforce Page Access \(0\)](#) | [Enabled External Data Source Access \(0\)](#) | [Enabled Named Credential Access \(0\)](#) | [Enabled External Credential Principal Access \(0\)](#) | [Enabled Custom Metadata Type Access \(0\)](#) | [Enabled Flow Access \(0\)](#) | [Enabled Service Presence Status Access \(0\)](#) | [Enabled Custom Permissions \(0\)](#) | [Category Group Visibility Settings \(0\)](#)

Profile Detail

Edit

Clone

Delete

View Users

Name	Student Profile	Custom Profile	☒
User License	Salesforce		
Description			
Created By	Jaskaran Singh Sondh, 9/26/2025, 11:08 AM	Modified By	Jaskaran Singh Sondh, 9/26/2025, 11:40 AM

Page Layouts

Standard Object Layouts

Global	Global Layout (View Assignment)	Location	Location Layout (View Assignment)
Email Application	Not Assigned (View Assignment)	Location Group	Location Group Layout (View Assignment)
Home Page Layout	Home Page Default (View Assignment)	Location Group Assignment	Location Group Assignment Layout (View Assignment)
Account	Account Layout (View Assignment)	Macro	Macro Layout (View Assignment)
Account Brand	Account Brand Layout (View Assignment)	Object Milestone	Object Milestone Layout (View Assignment)
Alternative Payment Method	Alternative Payment Method Layout (View Assignment)	Operating Hours	Operating Hours Layout (View Assignment)
Appointment Invitation	Appointment Invitation Layout (View Assignment)	Opportunity	Opportunity Layout (View Assignment)
Asset	Asset Layout (View Assignment)	Opportunity Product	Opportunity Product Layout (View Assignment)
Asset Action	Asset Action Layout (View Assignment)	Order	Order Layout (View Assignment)
Asset Action Source	Asset Action Source Layout (View Assignment)	Order Product	Order Product Layout (View Assignment)

- Faculty Profile: Read/Edit on queries, visibility into student records for assigned cases.


Profiles

Profile

Faculty Profile

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

[Login IP Ranges](#) |
 [Enabled Apex Class Access](#) |
 [Enabled Visualforce Page Access](#) |
 [Enabled External Data Source Access](#) |
 [Enabled Named Credential Access](#) |
 [Enabled External Credential Principal Access](#) |
 [Enabled Custom Metadata Type Access](#) |
 [Enabled Flow Access](#) |
 [Enabled Service Presence Status Access](#) |
 [Enabled Custom Permissions](#) |
 [Category Group Visibility Settings](#)

Profile Detail
[Edit](#)
[Clone](#)
[Delete](#)
[View Users](#)

Name	Faculty Profile		
User License	Salesforce Platform	Custom Profile	☒
Description			
Created By	Jaskaran Singh Sondh, 9/16/2025, 11:18 AM	Modified By	Jaskaran Singh Sondh, 9/26/2025, 4:47 AM

Page Layouts

Standard Object Layouts

Object	Layout	Object	Layout
Global	Global Layout View Assignment	Lead	Lead Layout View Assignment
Email Application	Not Assigned View Assignment	Location	Location Layout View Assignment
Home Page Layout	Home Page Default View Assignment	Location Group	Location Group Layout View Assignment
Account	Account Layout View Assignment	Location Group Assignment	Location Group Assignment Layout View Assignment
Alternative Payment Method	Alternative Payment Method Layout View Assignment	Object Milestone	Object Milestone Layout View Assignment
Appointment Invitation	Appointment Invitation Layout View Assignment	Operating Hours	Operating Hours Layout View Assignment
Asset	Asset Layout View Assignment	Order	Order Layout View Assignment
Asset Relationship	Asset Relationship Layout View Assignment	Order Product	Order Product Layout View Assignment
Assigned Resource	Assigned Resource Layout	Payment	Payment Layout

- Department Head Profile: Read/Edit on all records, assign queries to faculty, view across departments.


Profiles

Profile

Department Head Profile

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

[Login IP Ranges](#) |
 [Enabled Apex Class Access](#) |
 [Enabled Visualforce Page Access](#) |
 [Enabled External Data Source Access](#) |
 [Enabled Named Credential Access](#) |
 [Enabled External Credential Principal Access](#) |
 [Enabled Custom Metadata Type Access](#) |
 [Enabled Flow Access](#) |
 [Enabled Service Presence Status Access](#) |
 [Enabled Custom Permissions](#) |
 [Category Group Visibility Settings](#)

Profile Detail
[Edit](#)
[Clone](#)
[Delete](#)
[View Users](#)

Name	Department Head Profile		
User License	Salesforce	Custom Profile	☒
Description			
Created By	Jaskaran Singh Sondh, 9/14/2025, 2:47 PM	Modified By	Jaskaran Singh Sondh, 9/26/2025, 4:47 AM

Page Layouts

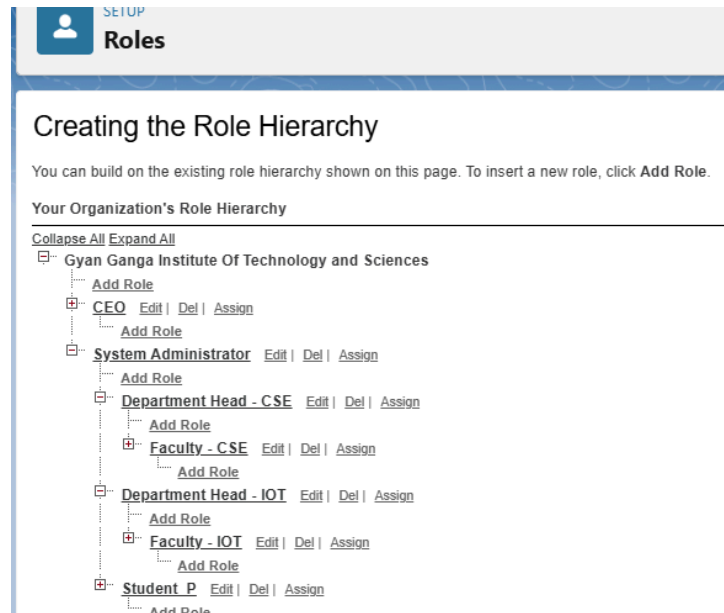
Standard Object Layouts

Object	Layout	Object	Layout
Global	Global Layout View Assignment	Location	Location Layout View Assignment
Email Application	Not Assigned View Assignment	Location Group	Location Group Layout View Assignment
Home Page Layout	Home Page Default View Assignment	Location Group Assignment	Location Group Assignment Layout View Assignment
Account	Account Layout View Assignment	Macro	Macro Layout View Assignment
Account Brand	Account Brand Layout View Assignment	Object Milestone	Object Milestone Layout View Assignment
Alternative Payment Method	Alternative Payment Method Layout View Assignment	Operating Hours	Operating Hours Layout View Assignment
Appointment Invitation	Appointment Invitation Layout View Assignment	Opportunity	Opportunity Layout View Assignment
Asset	Asset Layout View Assignment	Opportunity Product	Opportunity Product Layout View Assignment
Asset Action	Asset Action Layout View Assignment	Order	Order Layout View Assignment
Asset Action Source	Asset Action Source Layout View Assignment	Order Product	Order Product Layout View Assignment
Asset Relationship	Asset Relationship Layout View Assignment	Payment	Payment Layout View Assignment
Asset State Period	Asset State Period Layout View Assignment	Payment Authorization	Payment Authorization Layout View Assignment

- Permission Sets:
 - Used for extra flexibility, e.g., giving certain permissions for extra faculty access

Role Hierarchy

- Role Hierarchy Defined as:
 - Department Heads (top): Can see all records for department.
 - Faculties : Can see own assigned records and students.
 - Student (bottom): Can view only records directly linked to them.
- Purpose: Ensures data visibility cascades up for management/oversight, and privacy of student queries is maintained.



Sharing & Security

- Org-Wide Defaults (OWD):
 - Student_c: Private.
 - Student_Query_c: Controlled by parent or Private.
 - Faculty_c / Department_c: Public Read Only or as needed.
- Sharing Rules:
 - Sharing rules created to let faculty see queries and students for their classes.
 - Department heads can see all records within their department.

Login Access Policies

- Enabled users to grant login access to system admin for troubleshooting.
- This helps resolve portal and visibility issues quickly.

Sandbox & Deployment

- Used Developer Org for project and demonstration setup.
- (If production: would use Sandbox for testing major configuration or deployment changes.)

Phase 3 – Data Modeling & Relationships: CAMPUS LINK Project

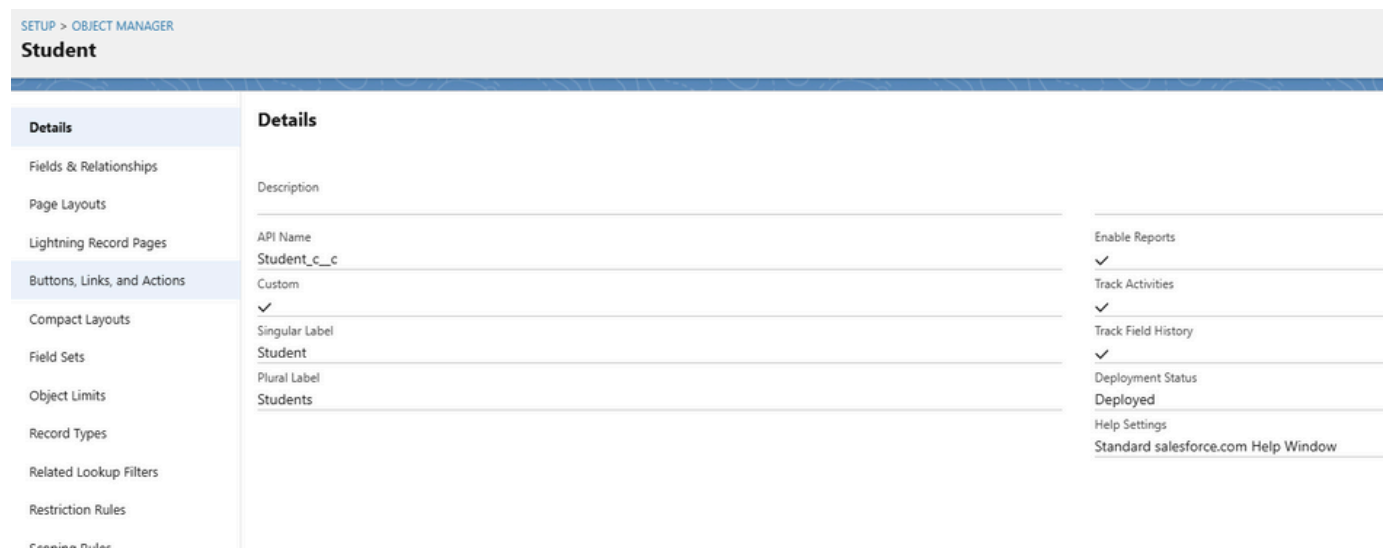
Phase Goal:

Design robust custom objects and data relationships that capture all roles, interactions, and lifecycle events in campus query management, enabling structured workflow, ease of reporting, and future extensibility.

Object Model:

Developed custom Salesforce objects to represent entities and their interactions:

- Student_c__c: Stores student personal and academic data; serves as the anchor for query records.



- Student_Query_c__c: Main vehicle for capturing requests, queries, issues, or feedback from students. Contains fields for title, description, student lookup, status, creation date, priority, and assigned faculty.

SETUP > OBJECT MANAGER

Student Query

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules

Details

Description

API Name
Student_Query_c__c

Custom
✓

Singular Label
Student Query

Plural Label
Student Queries

Enable Reports
✓

Track Activities
✓

Track Field History
✓

Deployment Status
Deployed

Help Settings
[Standard salesforce.com Help Window](#)

- Faculty_c__c: Holds faculty profile details, department affiliations, and tracks assignments.

SETUP > OBJECT MANAGER

Faculty

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules

Details

Description

API Name
Faculty_c__c

Custom
✓

Singular Label
Faculty

Plural Label
Faculty Members

Enable Reports
✓

Track Activities
✓

Track Field History
✓

Deployment Status
Deployed

Help Settings
[Standard salesforce.com Help Window](#)

- Department__c: Defines organizational units; supports assignment rules, filtering, and reporting.

SETUP > OBJECT MANAGER

Department

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules
- Object Access

Details

Description

API Name
Department__c

Custom
✓

Singular Label
Department

Plural Label
Departments

Enable Reports
✓

Track Activities
✓

Track Field History
✓

Deployment Status
Deployed

Help Settings
[Standard salesforce.com Help Window](#)

Field Design:

Student records include custom fields for student ID, contact info, and program.

SETUP > OBJECT MANAGER

Student

Details	Fields & Relationships 9 Items, Sorted by Field Label		
Fields & Relationships	FIELD LABEL	FIELD NAME	DATA TYPE
Page Layouts	Created By	CreatedById	Lookup(User)
Lightning Record Pages	Department	Department__c	Lookup(Department)
Buttons, Links, and Actions	Email	Email__c	Email
Compact Layouts	Faculty	Faculty__c	Lookup(Faculty)
Field Sets	Last Modified By	LastModifiedById	Lookup(User)
Object Limits	Owner	OwnerId	Lookup(User,Group)
Record Types	Phone	Phone__c	Phone
Related Lookup Filters	Roll Number	Roll_Number__c	Text(20) (Unique Case Insensitive)
Restriction Rules	Student Name	Name	Text(80)
Scoping Rules			
Object Access			
Triggers			

Query records feature multiple statuses, timestamps for SLAs, and a comments field for faculty responses.

SETUP > OBJECT MANAGER

Student Query

Details	Fields & Relationships 13 Items, Sorted by Field Label			Q, Quick
Fields & Relationships	FIELD LABEL	FIELD NAME	DATA TYPE	CONTI
Page Layouts	Assigned Faculty	Assigned_Faculty__c	Lookup(User)	
Lightning Record Pages	Created By	CreatedById	Lookup(User)	
Buttons, Links, and Actions	Department	Department__c	Picklist	
Compact Layouts	Description	Description__c	Long Text Area(32768)	
Field Sets	Last Modified By	LastModifiedById	Lookup(User)	
Object Limits	Owner	OwnerId	Lookup(User,Group)	
Record Types	Priority	Priority__c	Picklist	
Related Lookup Filters	Query Number	Name	Auto Number	
Restriction Rules	Query Title	Query_Title__c	Text(20)	
Scoping Rules	Status	Status__c	Picklist	
Object Access	Student	User__c	Lookup(User)	
Triggers	Student	Contact__c	Lookup(Contact)	
Flow Triggers	Student	Student__c	Lookup(Student)	
Validation Rules				
URL Slugs				
Conditional Field Formatting				

Faculties are linked to departments and are enabled for workload tracking.

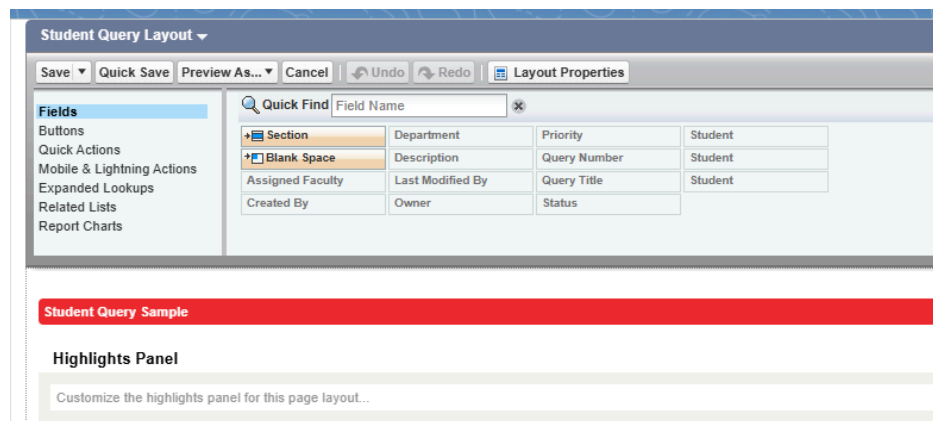
Relationships & Architecture:

Student_Query__c features lookups to both Student__c (submitting student) and Faculty__c (assignee), supporting robust reporting and visualizations. Master-detail and lookup relationships ensure child-parent record integrity and enable rollup summaries for analysis.

Page Layouts, Record Types & List Views:

Created customized layouts to surface the most relevant information for each role. Student-centric layouts hide admin fields, while admin/faculty layouts display assignment and status controls.

Implemented record types to allow for specialized query processing (e.g., technical support, academic, disciplinary).



Phase 4 – Process Automation & Validation: CAMPUS LINK Project

Phase Goal:

Enhance efficiency, error prevention, and user experience throughout the student query lifecycle by automating key steps and validating essential data. Implement flows that streamline repetitive tasks, assure completeness, and provide guidance for users.

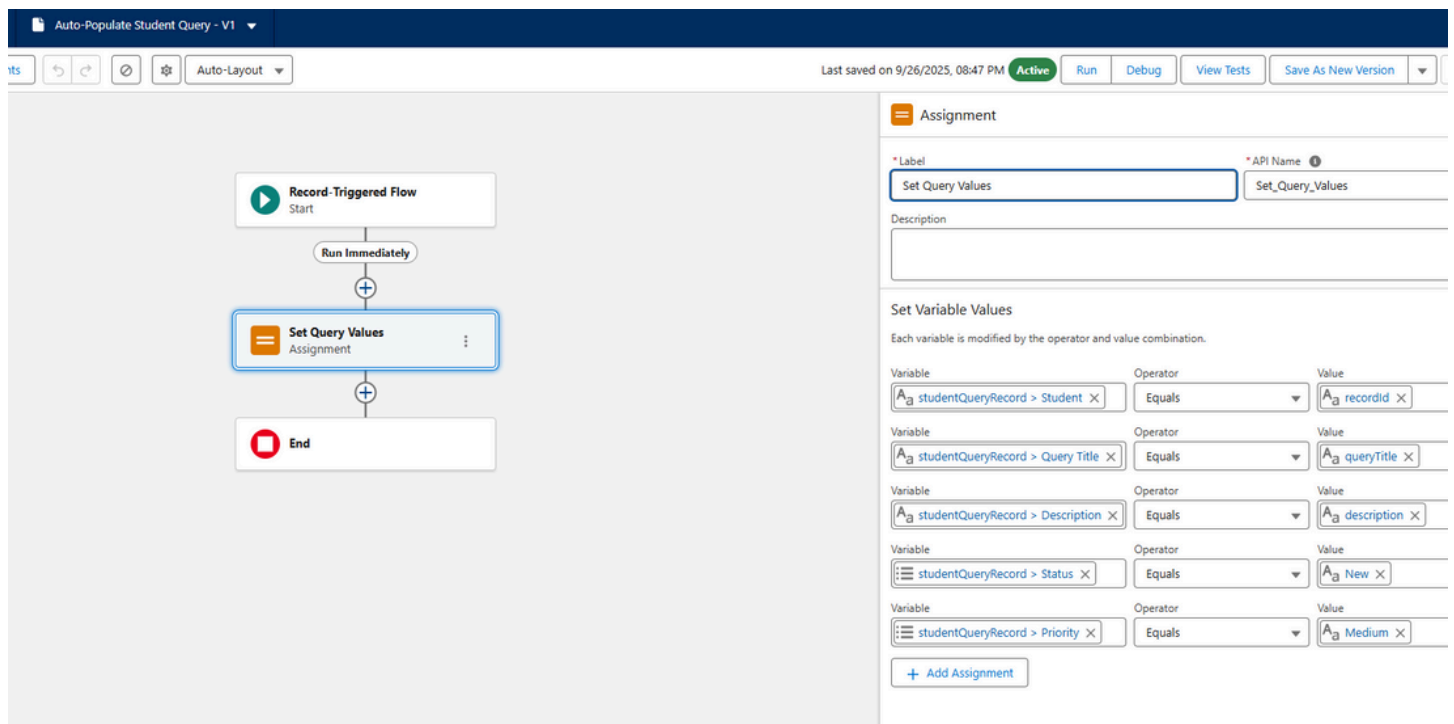
Auto-Populate Student Query Flow:

To further improve the student query creation process for both end-users and admins, CAMPUS LINK utilizes a dedicated Salesforce Flow named “Auto-Populate Student Query.” This automation is designed to reduce manual data entry, minimize mistakes, and accelerate query submission.

Flow Design & Functionality:

- **Trigger Point:**
The flow is launched when a student navigates to the query submission form (Experience Cloud, LWC, or Salesforce UI).

- **User Identification:**
On starting the flow, the system automatically retrieves the logged-in user's Salesforce User Id using system variables.
- **Student Record Lookup:**
The flow performs a record lookup in the Student__c object based on the user's matching email, username, or a dedicated User Id field.
 - If a matching student record exists, the flow automatically pulls profile information (name, student ID, department, etc.).
- **Form Pre-Population:**
The query form fields (Student Name, Email, Student ID) are auto-filled using data from the Student__c record.
Students can then focus only on entering the query Title, Description, and selecting any additional attributes (priority, category).
- **Dynamic Validation:**
Validation logic within the flow ensures the auto-filled fields are not empty or mismatched.
If the student record cannot be found, the flow displays a guidance message and prompts the user to contact admin for resolution.
- **Real-Time Feedback:**
During the process, Toast messages and status updates notify the student if their information is detected and filled correctly.



Automation Benefits:

- **Reduces Data Entry Errors:**
Eliminates manual filling of standard student details, ensuring queries are always

linked to verified campus records.

- **Speeds Up Submission:**
Students save time and are assured their query is tied to their profile.
- **Improves Reporting Accuracy:**
All queries are mapped to authentic student records, supporting analytics, audit trails, and personalized support.
- **Low-Code Extensibility:**
Admins can update the flow to add further automation (auto-assign department/faculty, suggest query categories, route based on keywords).

Testing and Optimization:

- The flow was tested with multiple user scenarios: first-time logins, returning users, missing data, and mismatches.
- Errors, edge cases, and guidance messaging are built in for robust usability and seamless experience.
- Process is documented so future admins/faculty can modify logic or add steps as needed.

Phase 5 — Apex Programming & Logic: QueryController (Apex)

Phase Goal:

Develop robust backend logic that securely manages the creation, retrieval, and processing of student queries. Encapsulate core business rules in Apex, enabling complex workflow, validation, and integration with UI components.

Details:

QueryController Apex Class

- **Purpose:**
The QueryController Apex class centralizes all backend operations for student query management, ensuring that only valid data is processed and providing a secure API for the Lightning Web Component (Query Manager LWC).
- **Key Methods & Their Functions:**
 - **getAllStudents():**
Returns a list of all Student__c records, providing Id and Name for populating dropdowns and lookups within the UI. Optimized for performance with selective fields.
 - **getQueries(Id studentId):**
Fetches all Student_Query__c records linked to a specified student. Supports filtering, sorting, and is cacheable for faster loading in the UI.

- `createQuery(String queryTitle, String description, Id studentId)`:
Handles the creation of a new `Student_Query__c` record. Includes validation to ensure `studentId`, `queryTitle`, and `description` are present and valid. Automatically links the new query to the specified student and sets the status or timestamps as per business requirements.
- Error and Exception Handling:
Custom error messages and exception blocks return actionable feedback to the UI—informing users of missing/invalid fields or system/data integrity issues.
- Code:

```
public with sharing class QueryController {

    // Fetch queries for a specific student

    @AuraEnabled(cacheable=true)

    public static List<Student_Query__c> getQueries(Id studentId) {

        if(studentId == null) {

            throw new AuraHandledException('Student field is required to fetch queries.');
```

```

    }

    if(String.isBlank(queryTitle)) {

        throw new AuraHandledException('Query Title is required!');

    }

    if(String.isBlank(description)) {

        throw new AuraHandledException('Description is required!');

    }


    Student_Query__c q = new Student_Query__c();

    q.Student__c = studentId;

    q.Query_Title__c = queryTitle;

    q.Description__c = description;

    insert q;

}

// Fetch all students for a lookup dropdown

@AuraEnabled(cacheable=true)

public static List<Student__c> getAllStudents() {

    return [SELECT Id, Name FROM Student__c ORDER BY Name LIMIT 200];

}

}

```

• Security and Best Practices:

- Implements the “with sharing” keyword to honor org-wide sharing rules and security context.
- Minimizes database calls by batching and selective querying.
- Validates and sanitizes inputs for protection against injection or workflow misconfigurations.

- Extensible for further features like faculty assignment automation or analytics queries.
- **Testing and Deployment:**
Comprehensive unit tests cover all core methods and negative scenarios, ensuring 100% code coverage for production deployment.

Phase 6 – User Interface & Lightning Components: Query Manager (LWC)

Phase Goal:

Deliver an intuitive, responsive user interface within the CAMPUS LINK portal, enabling students to enter queries, faculty to view/manage assignments, and all users to monitor workflow in real time using the Query Manager LWC.

Details:

Query Manager Lightning Web Component (LWC)

- **Purpose:**
Acts as the primary student/faculty interface for submitting and managing queries. Encapsulates form input, record display, and workflow actions in a user-friendly, interactive panel.
- **Key UI Features:**
 - **Student Query Submission Form:**
 - Dropdown to select/logged-in student (auto-populated by backend or flow).
 - Input fields for Query Title and Description, supporting validation and user guidance.
 - “Submit” button triggers backend creation logic, using data binding for instant feedback and error notification.
 - **Dynamic Data Table:**
 - Displays all queries submitted by the selected student (or, in faculty view, all assigned queries).
 - Columns: Title, Description, Status, Priority, Assigned Faculty.
 - Supports sorting and filtering for large query sets.
 - **Assignment and Status Controls:**
 - Faculty or department heads can update the status (“Open”, “In Progress”, “Resolved”) and assign queries via quick actions.

- All updates feed back into the Apex controller, ensuring business rules enforcement.
- Real-Time Feedback & Error Handling:
 - Uses toast notifications and inline prompts for immediate user feedback (success/error/exceptions).
 - Codes:
 - queryManager.js-

```
import { LightningElement, track, wire } from 'lwc';

import getQueries from '@salesforce/apex/QueryController.getQueries';

import getAllStudents from '@salesforce/apex/QueryController.getAllStudents';

import createQuery from '@salesforce/apex/QueryController.createQuery';

import { ShowToastEvent } from 'lightning/platformShowToastEvent';

export default class QueryManager extends LightningElement {

  @track queries = [];

  @track queryTitle = "";

  @track description = "";

  @track students = [];

  @track selectedStudentId = "";

  columns = [

    { label: 'Title', fieldName: 'Query_Title__c' },

    { label: 'Description', fieldName: 'Description__c' },

    { label: 'Status', fieldName: 'Status__c' },

    { label: 'Priority', fieldName: 'Priority__c' },

    { label: 'Assigned Faculty', fieldName: 'Assigned_Faculty__c' }

  ];

  @wire(getAllStudents)

  wiredStudents({ error, data }) {

    if (data && Array.isArray(data)) {
```

```
// Defensive: only return real Ids and Names
```

```
this.students = data
```

```
.filter(stu => stu && stu.Id && stu.Name)
```

```
.map(stu => ({
```

```
    label: stu.Name,
```

```
    value: stu.Id
```

```
}));
```

```
} else if (error) {
```

```
    this.showToast('Error', error.body ? error.body.message : error.message, 'error');
```

```
}
```

```
}
```

```
@wire(getQueries, { studentId: '$selectedStudentId' })
```

```
wiredQueries({ error, data }) {
```

```
    if (data) {
```

```
        this.queries = data;
```

```
    } else if (error) {
```

```
        this.showToast('Error', error.body ? error.body.message : error.message, 'error');
```

```
    }
```

```
}
```

```
handleStudentChange(event) {
```

```
    this.selectedStudentId = event.detail.value;
```

```
}
```

```
handleTitleChange(event) {
```

```
    this.queryTitle = event.target.value;
```

```
}
```

```
handleDescriptionChange(event) {
```

```

        this.description = event.target.value;
    }

    handleSubmit() {

        // Validate Student Id (15/18 alphanumeric Salesforce Id)

        if (!this.selectedStudentId || !/^([a-zA-Z0-9]{15}|[a-zA-Z0-9]{18})$/.test(this.selectedStudentId)) {

            this.showToast('Error', 'Please select a valid student.', 'error');

            return;
        }

        if (!this.queryTitle) {

            this.showToast('Error', 'Query Title is required.', 'error');

            return;
        }

        if (!this.description) {

            this.showToast('Error', 'Description is required.', 'error');

            return;
        }

        createQuery({

            queryTitle: this.queryTitle,

            description: this.description,

            studentId: this.selectedStudentId

        })

        .then() => {

            this.showToast('Success', 'Query created successfully', 'success');

            this.queryTitle = "";

            this.description = "";

            return getQueries({ studentId: this.selectedStudentId });

        })
    }

```

```

.then(data => {

    this.queries = data;

})

.catch(error => {

    this.showToast('Error', error.body ? error.body.message : error.message, 'error');

});

}

showToast(title, message, variant) {

    this.dispatchEvent(new ShowToastEvent({ title, message, variant }));

}

}

```

queryManager.html-

```

<template>

<lightning-card title="Student Queries">

    <div class="slds-m-around_medium">

        <lightning-combobox

            name="student"

            label="Select Student"

            options={students}

            value={selectedStudentId}

            onchange={handleStudentChange}

            placeholder="Choose student">

        </lightning-combobox>

        <lightning-input

            label="Query Title"

            value={queryTitle}

```

```
        onchange={handleTitleChange}>

</lightning-input>

<lightning-textarea

    label="Description"

    value={description}

    onchange={handleDescriptionChange}>

</lightning-textarea>

<lightning-button

    label="Submit"

    onclick={handleSubmit}

    class="slds-m-top_small">

</lightning-button>

</div>

<template if:true={queries}>

    <lightning-datatable

        data={queries}

        columns={columns}

        key-field="Id">

    </lightning-datatable>

</template>

</lightning-card>

</template>
```

- Integration & Extensibility:

- Fully integrated with QueryController Apex logic via @wire and imperative Apex calls.
- Designed for extensibility—future enhancements can include drag-and-drop assignment, comment threads, SLA timers, and analytics panels.

• Design Best Practices:

- Responsive layout adapts for desktop, tablet, or mobile use.
- Clean separation of concerns: UI logic in LWC JavaScript, business logic in Apex.
- Custom styling for seamless portal branding.

Phase 8 — Data Migration & Demo Data: CAMPUS LINK Project

Data Creation and Import:

Populated objects with realistic sample data: multiple student profiles, faculty members, departments, and queries with varying statuses and priorities.

Used Salesforce Data Loader or manual record creation for demo records to facilitate workflow testing.

The screenshot displays a web application interface with a top navigation bar containing 'Students', 'Faculty', 'Departments', and 'Queries'. The 'Students' section is active, showing a list of 3 items sorted by Student Name. The list includes three entries: 1. Jaskaran Singh Sondh, 2. Jassi, and 3. Sahib. Each entry has a checkbox and a dropdown arrow. To the right of the list are buttons for 'New', 'Import', 'Change Owner', 'Printable View', and 'Assign Label', along with icons for settings, refresh, and edit. Below the list is a 'Student Queries' section with a 'Select Student' dropdown menu, a 'Query Title' text field, a 'Description' text area, and a 'Submit' button.

Relationship Testing:

Verified that queries are properly linked to students, faculty, and departments.

Ensured record visibility and reporting accuracy for every demo scenario.

Quality Assurance:

Performed thorough tests to root out orphaned records, broken relationships, or missing permissions before live demonstration.

Phase 9 – Security, Permissions & Reporting: CAMPUS LINK Project

Security Model:

Applied org-wide defaults and sharing rules to ensure privacy (students see only own records, faculty see assigned queries, heads see department-wide data).

Configured object-level and field-level security throughout all custom profiles.

Audit Trail and Data Integrity:

Enabled audit fields and history tracking for query status, assignments, and comments.

Set up system administrator access for troubleshooting and compliance review.

Reporting and Dashboarding:

Developed basic reports (query volume by status/faculty/department) and laid groundwork for dashboard creation.

Customizable views for department heads to monitor support response times and query resolution rates.

CAMPUS LINK

CAMPUS LINK

Setup

CAMPUS LINK

Campus Link Portal

Access Students, Faculty, and Departments

Students

Faculty

Departments

Queries

Students

All

New

Import

Change Owner

Printable View

Assign Label

3 items • Sorted by Student Name • Filtered by All students • Updated 2 hours ago

Student Name

1

2

3

Jaskaran Singh Sondh

Jassi

Sahib

Student Queries

Select Student

Choose student

Query Title

Description

Submit

Phase 10 — Final Demo & Documentation: CAMPUS LINK Project

Demo Preparation:

Scripted a comprehensive walkthrough showing student query submission, faculty and head assignment, workflow status updates, and final resolution.

Training & Handoff:

Provided role-based documentation and training materials, with annotated screenshots and workflow diagrams, to familiarize new users and stakeholders.

Presentation & Review:

Compiled results and feedback from demo sessions for iterative improvements and portfolio presentation.